

Ahmed Ikram | AI Engineer

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Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee (predicted First, graduating June 2027). Built LLM systems with the evaluation harnesses that prove they work: RAGAS-scored retrieval, QLoRA and vLLM serving. Seeking a graduate AI engineering role from June 2027.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a **First-Class Honours** degree, Graduation: **June 2027**

SKILLS

Languages & Data: Python · SQL · TypeScript · JavaScript · Bash · Rust / PyO3 · PostgreSQL · Redis · Kafka · Pydantic

LLMs: PyTorch · Hugging Face Transformers · QLoRA/LoRA/PEFT · Unsloth · TRL · Accelerate · BitsandBytes · vLLM · Jinja 2, Prompt & Context Engineering

Agents & RAG: LangGraph · LangChain · Model Context Protocol (MCP) · AWS Bedrock · ChromaDB · Cross-Encoder Reranking · Multimodal (Vision-Language) Extraction · Guardrails

Evaluation & Observability: RAGAS · Execution-Based Scoring · LLM-as-Judge Evaluation · LangSmith · Langfuse · Evidently AI

MLOps & Serving: MLflow · Weights & Biases · Apache Airflow · FastAPI · Modal · AWS SageMaker

Classic ML: XGBoost · Isolation Forest · scikit-learn · Optuna · Tesseract · OpenCV

Cloud: AWS · Google Cloud (BigQuery) · Terraform · Docker · GitHub Actions · Linux

PROJECTS

SWE-Qwen Automated LLM Fine-Tuning, Evaluation & Serving

Python · PyTorch · QLoRA · Unsloth · Hugging Face · Modal · vLLM · Weights & Biases · Langfuse · Terraform · Google Cloud

- Model-agnostic LLMops platform; schema-validated data preparation, QLoRA fine-tuning, execution-based evaluation and statistically-gated champion/challenger promotion, serving the winner as a config-driven OpenAI compatible API - versioned GCS/W&B artifacts shared across every stage.
- Six-stage data pipeline turned 20,477 raw SWE-bench issues into a leak-free 17,456-example corpus across 46 repos; evaluation materialises each repo in its official Docker image and runs its real pass/fail test suite for ground-truth, execution-based scoring.
- Scored 3 QLoRA Qwen3-14B variants (4-bit NF4, Unsloth + FlashAttention on Modal A100); champion reached 17.20% fail-to-pass - 7.0× base (95% CI 11.1-25.8%, McNemar $p < 1e-6$) - and 90.10% unrelated pass-pass rate (+61.6 pt vs base); Promotion requires a 4-condition statistical gate so a challenger can never self-certify a win.
- Served on Modal scale-to-zero with per-request LoRA swapping via vLLM+AWQ; a 20-instance F2P smoke gate (SMOKE_TOLERANCE=0.05) blocks regressing PRs. GCP via Terraform + Workload Identity Federation; 1,456 tests.

ATM Log Aggregation, Anomaly Detection & Diagnostics - RAG Assistant and ML Anomaly Detection (NCR Atleos)

LangGraph · MCP · ChromaDB · FastAPI · XGBoost · MLflow · Kafka · PostgreSQL · Redis · Docker · Terraform · AWS

- Hybrid RAG as a LangGraph agent over a 12-tool MCP server, semantic chunking + local `nomic-embed-text` embeddings, cross-encoder reranking, grounding-gated re-retrieval, then reflexion / self-consistency / citation checks fused with Platt-calibrated confidence (ECE gate 0.10).
- RAGAS evaluation across 3 retrieval strategies (50-query golden set): adaptive routing with re-retrieval nearly doubles context recall over deterministic selection (0.708 vs 0.350) at the best context precision (0.874); a CI gate blocks any merge regressing a RAGAS metric by >0.05 .
- 3-layer anomaly detection (XGBoost + Isolation Forest ensemble, Z-score & 7-heuristics engine) runs every 30s on a live Kafka stream, reaching 0.70 F1 at 97.3% precision via a 200-point Youden sweep, versioned in MLflow with champion aliases and cross-checked against a live SageMaker endpoint.
- Terraform-provisioned AWS (ECS Fargate, RDS, S3, SageMaker, Secrets Manager) via OIDC-authenticated CI/CD, validated by 1,438 tests.

StockLens FinTech Spending & Investment App - AI Agent, Multimodal Extraction, MLOps

LangGraph · MCP · PyTorch · Optuna · Rust / PyO3 · Airflow · MLflow · Evidently AI · Tesseract · PostgreSQL · Redis · Terraform

- 2-node LangGraph ReAct agent (StateGraph, not `create\_react\_agent` wrapper) exposes 16 tools to AWS Bedrock over a FastAPI-mounted MCP server (Streamable HTTP, OAuth 2.1 PKCE). Correctness judged in CI by an LLM-as-judge with LangSmith tracing.
- Confidence-gated receipt-extraction cascade (Tesseract OCR → Bedrock Vision-LLM → Text fallback) with rapidfuzz merchant matching, escalating only when confidence drops below thresholds.
- Global LSTM (entity embeddings, Optuna 50-trial HPO, focal loss) reaches 51.6% 5-day directional accuracy vs a 33% 3-class baseline (0.97 simulated Sharpe). 17 features computed in a Rust/PyO3 engine; retrained weekly via Airflow with Evidently drift monitoring (PSI/KS/JSD) and MLflow champion/challenger promotion at >2 pp.
- 2,304 tests with a 90% test coverage gate on Terraform-provisioned AWS (ECS Fargate, EFS champion serving); 9-parallel CI jobs and a 7-stage OIDC-authenticated CD.

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>

EXPERIENCE

Beaully Mini Market, Dundee, Scotland

Retail Assistant

September 2023 - Present

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker

June 2021 - September 2023